C 42790 **(Pages : 5)** **Name.....**

Reg. No.....

SECOND SEMESTER M.Sc. DEGREE (REGULAR/SUPPLEMENTARY) EXAMINATION, APRIL 2023

(CBCSS)

Mathematics

MTH 2C 09—ODE AND CALCULUS OF VARIATIONS

(2019 Admission onwards)

Time : Three Hours Maximum : 30 Weightage

Part A

*Answer **all** questions.*
Each question carries 1 *weightage.*

1. Give confluent hypergeometric equation.

2. Find the first three terms of the Legendre series of

$$f(x)=\begin{cases}0 & \text{if} \quad -1\le x<0,\\ x & \text{if} \quad 0\le x\le 1.\end{cases}$$

3. Determine whether the following functions if positive definite, negative definite, or neither :

$$-2x^2+3xy-y^2.$$

4. Show that $(0,0)$ is a simple critical point of the system :

$$\begin{cases}\dfrac{dx}{dt}=-2x+3y+xy\\ \dfrac{dy}{dt}=-x+y-2xy^2\end{cases}.$$

5. Describe the phase portrait of the system

$$\begin{cases}\dfrac{dx}{dt}=x\\ \dfrac{dy}{dt}=0\end{cases}.$$

Turn over

6. Show that $f(x, y) = xy^2$ satisfies a Lipschitz condition on any rectangle $a \le x \le b$ and $c \le y \le d$.

7. Describe Picard's iteration method.

8. Give the solution, if exists, to the initial value problem

$$y' = t, y(0) = 1.$$

(8 × 1 = 8 weightage)

Part B

Answer any **two** *questions from each of the following* **three** *units.*
Each question carries 2 *weightage.*

UNIT I

9. Express $\sin^{-1} x$ in the form of a power series by solving the equation

$$y' = \left(1 - x^2\right)^{-1/2}, y(0) = 0$$

in two ways.

10. Locate and classify singular points on the x-axis of the differential equation

$$x^2\left(x^2 - 1\right)^2 y'' - x(1 - x)y' + 2y = 0.$$

11. Find the general solution of the differential equation

$$\left(2x^2 + 2x\right)y'' + (1 + 5x)y' + y = 0$$

near the singular point $x = 0$.

UNIT II

12. If the two solutions

$$\begin{cases} x = x_1(t) \\ y = y_1(t) \end{cases} \quad \text{and} \quad \begin{cases} x = x_2(t) \\ y = y_2(t) \end{cases}$$

of the homogeneous system

$$\begin{cases} \dfrac{dx}{dt} = a_1(t)\,x + b_1(t)\,y \\ \dfrac{dy}{dt} = a_2(t)\,x + b_2(t)\,y \end{cases}$$

are linearly independent on $[a, b]$, then prove that

$$\begin{cases} x = c_1 x_1(t) + c_2 x_2(t) \\ y = c_1 y_1(t) + c_2 y_2(t) \end{cases}$$

is the general solution of the homogeneous system on this interval.

13. Prove that the function $E(x, y) = ax^2 + bxy + cy^2$ is positive definite if and only if $a > 0$ and $b^2 - 4ac < 0$, and is negative definite if and only if $a < 0$ and $b^2 - 4ac < 0$.

14. Determine the nature and stability properties of the critical point $(0, 0)$ for the following linear autonomous system :

$$\begin{cases} \dfrac{dx}{dt} = 4x - 3y \\ \dfrac{dy}{dt} = 8x - 6y \end{cases}.$$

UNIT III

15. Let $y_p(x)$ be a non-trivial solution of Bessel's equation on the positive x-axis. If $0 \le p < 1/2$, then prove that every interval of length π contains at least one zero of $y_p(x)$; if $p = 1/2$, then prove that the distance between successive zeros of $y_p(x)$ is exactly π; and if $p > 1/2$, then prove that every interval of length π contains at most one zero of $y_p(x)$.

16. Given $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. By Picard's iteration method, find approximate value of y for $x = 0.2$ and $x = 1$.

17. Prove that the geodesics on a sphere are arcs of great circles.

(6 × 2 = 12 weightage)

Part C

Answer any **two** *questions.*
Each question carries 5 *weightage.*

18. (a) Find a power series solution of the form $\sum a_n x^n$ of the differential equation

$$y' + y = 1.$$

(b) Show that equation

$$4x^2y'' - 8x^2y' + \left(4x^2 + 1\right)y = 0$$

has only one Frobenius series solution and find it.

19. (a) Derive Rodrigue's formula for Legendre polynomials

$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}\left(x^2 - 1\right)^n.$$

(b) If the roots m_1 and m_2 of

$$m^2 - (a_1 + b_2)\, m + (a_1 b_2 - a_2 b_1) = 0$$

are real, distinct, and of the same sign, then prove that the critical point $(0, 0)$ of the system

$$\begin{cases} \dfrac{dx}{dt} = a_1 x + b_1 y \\ \dfrac{dy}{dt} = a_2 x + b_2 y \end{cases}$$

is a node.

20. (a) If $q(x) < 0$, and if $u(x)$ is a nontrivial solution of

$$u'' + q(x)\, u = 0$$

then prove that $u(x)$ has at most one zero.

(b) For the following nonlinear system :

i) Find the critical points ;

ii) Find the differential equation of the paths ; and

iii) Solve this equation to find the paths

$$\begin{cases} \dfrac{dx}{dt} = e^y \\ \dfrac{dy}{dt} = e^y \cos x \end{cases}$$

21. (a) Find the curve joining two points (x_1, y_1) and (x_2, y_2) that yields a surface of revolution of minimum area when revolved about the x-axis.

(b) State and prove Picard's theorem.

(2 × 5 = 10 weightage)